

3(a)(i)	<i>any three from</i> (polymer / polysaccharide of) β-glucose ; glycosidic, bonds / linkages ; A glucosidic (β) 1–4 (glycosidic) bonds ; R if 1–6 also given <i>ref.</i> to (β) glucose units, linked at 180° to each other / alternately orientated / AW ; unbranched (polymer) / straight chain / linear ; able to form hydrogen bonds with parallel chains / AW ;	3
3(a)(ii)	chloroplast / granum / thylakoid / middle lamella / large vacuole / nucleus at edge of cell ;	1
3(a)(iii)	provide energy (for the cell) ; R produce energy any two examples ;; e.g. biosynthesis A named example active transport / proton pumping movement / described, e.g. movement of, vesicles / chromosomes endocytosis / exocytosis	3
3(b)	<i>any two from</i> water molecules are dipoles ; A polar (each water molecule has) δ^+ hydrogen (atoms) and a δ^- oxygen (atom) ; the positively charged hydrogen (atom) of one water molecule is attracted to the negatively charged oxygen (atom) of another water molecule ; weak attraction between water molecules ;	2
3(c)	<i>any two from</i> water molecules are polar ; ions are charged ; <i>ref.</i> attraction between water molecules and ions ; AVP ; e.g. oxygen δ^- faces positive ion	2